

CG IP Project in Action

The "Repo and Hello World pipeline" setup

Objective: To ensure every team member completes the session with a working, unified development environment - a shared GitHub repository, and a running baseline code pipeline.

Step 1. Repository and branch setup

Task - The team lead creates a shared **GitHub repository** and invites all members.

Setup - Create a standard directory structure:

```
├─ docs/
├─ src/
│   └─ gui/
│       └─ processing/
│           └─ main.py
├─ tests/
├─ requirements.txt
└─ README.md
```

Git strategy: Configure branch protection rules for **main** and create individual feature branches (e.g., **feature/ui-setup**, **feature/image-loader**).

Step 2. Environment alignment and dependencies

Task - Create a virtual environment (**venv** or **conda**) on laptop of every team member using the technology stack selected in Part B.

Action - Generate a **requirements.txt** file containing core libraries specified in your proposal.

```
opencv-python
PyOpenGL
PyQt5 # or tkinter / customtkinter
numpy
matplotlib
```

Verification - Run **pip install -r requirements.txt** on all team members' laptops to ensure identical package versions.

Step 3. The "Hello World" pipeline test

Task - Write a minimal Python script (`main.py`) that executes a end-to-end check connecting their GUI, Image Processing, and Visualization modules.

Sample 15-line smoke test code:

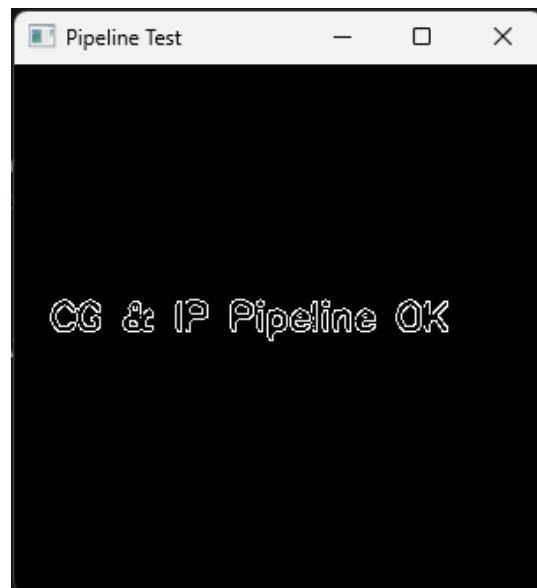
```
import cv2
import numpy as np

# Load dummy or sample image
img = np.zeros((300, 300, 3), dtype=np.uint8)
cv2.putText(img, 'CG & IP Pipeline OK', (20, 150),
            cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0), 2)

# Apply a basic baseline operation
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray, 100, 200)

# Display to confirm GUI window rendering
cv2.imshow('Pipeline Test', edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Expected Output



Step 4. Initial PR & First Commit

Task - Every member pushes their initial branch, opens a Pull Request (PR), and performs a cross-peer code review before merging into `main`.

Deliverable submission: The team leader posts repo link of group and a screenshot of the baseline test running on his/her laptop in our WhatsApp group.